MIL-E-44166A(GL)
22 December 1986
SUPERSEDING
MIL-E-44166(GL)
29 March 1985

MILITARY SPECIFICATION

EXTRACTOR, LAUNDRY: HEAVY DUTY, TOP LOADING, TRAILER MOUNTED

This specification is approved for use by the U.S. Army Natick Research, Development, and Engineering Center, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers a trailer mounted laundry extractor for field use (see 6.1)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

HH-I-553 - Insulation Tape, Electrical (Rubber, Natural and Synthetic)
QQ-S-571 - Solder, Tin Alloy: Tin, Lead Alloy; and Lead Alloy

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improving Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3510

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MILITARY

MIL-T-152	-	Treatment, Moisture and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment
MIL-T-704	-	Treatment and Painting of Material
MIL-L-3153	-	Laundry and Dry Cleaning Machinery and Equipment Packaging of
MIL-T-44047	-	Trousers, Camouflage Pattern; Combat
MIL-L-44142	-	Laundry Unit, Trailer Mounted

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-195	-	Marking of Connections for Electrical Assemblies
MIL-STD-810	-	Environmental Test Methods and Engineering Guide Lines

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

6-1-9945	-	Extractor, Laundry, Trailer Mounted Assembly

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issues of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA No. 70 - National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02169.)

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NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

MG 1 - Motors and Generators

(Application for NEMA standards should be addressed to the National Electrical Manufacturers Association, 2101 L St., N.W., WAshington, DC 20037.)

AMERICAN WELDING SOCIETY (AWS)

Standard Qualification Procedure of the American Welding Society

(Application for copies should be addressed to the American Welding Society, 345 East 47th St., New York, NY 10017.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Welding Qualifications of the American Society of Mechanical Engineers

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th St., New York, NY 10017.)

UNDERWRITERS LABORATORIES INC.(UL)

UL 50 - Cabinets and Boxes

(Application for copies should be addressed to Underwriters Laboratories Inc., 333 Pfingsten Rd., Northbrook, IL 60062.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 240 - Heat-Resisting Chromium and Chromium-Nickel Stainless Steel Plate, Sheet and Strip for Fusion-Welded Unfired Pressure Vessels

B 584 - Casting, Copper Alloy Sand, for General Application

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the reference cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Codes and standards. The laundry extractor shall comply with the applicable requirements of National Fire Protection Association NFPA 70, Underwriters Laboratories Inc., UL 50, and National Electrical Manufacturers Association MG 1. Satisfactory evidence that these requirements have been met shall be submitted to the contracting officer or his authorized representative prior to the start of production (see 4.5).

3.3 Materials and components. The materials and components shall be as specified herein and on the applicable drawings. Materials and components not specifically defined shall be of the quality and construction required to meet the laundry extractor's intended use provided the completed item complies with all of the provisions of this document (see 6.4). All extractors, control units, and related parts furnished under any specific contract shall be physically and mechanically identical. This requirement includes parts, assemblies, components and accessories. No changes or deviations shall be acceptable without written approval of the contracting officer.

3.3.1 Basket sides, top and bottom, lid, curb, and belt guard. The basket sides, top and bottom, lid, curb, and belt guard shall be stainless steel conforming to ASTM A 240.

3.3.2 Basket centerpost. The basket centerpost shall be a copper alloy sand casting conforming to UNS 97600 of ASTM B 584. The centerpost shall be cast around a stamped steel insert that is permanently imbedded in the casting. The steel insert will be designed to contact the drive mechanism on the drive shaft to ensure proper operation.

3.3.3 Pullies, legs, and drive unit housings. The pullies, legs and drive unit housings shall be constructed from cast iron.

3.3.4 Drive shaft. The drive shaft shall be constructed out of high strength medium carbon (0.40 to 0.50) steel with a yield strength of 100,000 psi and Brinell hardness of 269 plus or minus 2 percent.

3.4 Alternate materials and components. Materials and components offered as equivalents to those specified herein and on the applicable drawings shall be equal in form, fit and function to the specified material or component. The incorporation of such a material or component material in the design of the specified laundry extractor shall not require modification or change in location of any other material or component material, unless such modification is specifically approved in writing by the contracting officer. The contractor shall

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submit with his bid for the contracting officer's or contracting officer's representative approval, a list identifying each alternate material or component, that each listed component complies with the requirements listed herein. The contracting officer, may require a physical sample of the proposed substitution. Approval of the submitted listing, together with the necessary supporting data does not relieve the contractor of the responsibility that these materials or components perform in accordance with specified requirements when incorporated into the trailer mounted laundry extractor.

3.5 Design and construction. Design and construction of the trailer mounted laundry extractor shall be as specified herein and on Drawing 6-1-9945 and all subsidiary drawings and parts lists pertaining thereto.

3.5.1 Extractor. The extractor shall be heavy duty top loading capable of processing not less than 50 pounds dry weight of material per cycle. All components shall be fully accessible including an externally mounted control panel and motor. The rotation of the basket shall not be less than 1700 revolutions per minute. The extractor shall be mounted and operated on an Army M979 tandem axle trailer.

3.5.1.1 Basket. The basket shall be 24 inches in diameter by 14.25 inches deep. The perforations in the basket shall face upward and be formed without removing any of the sidewall or bottom material. The basket shall be integral with a minimum 1 inch thick steel flywheel which is joined to the basket by riveting or welding. The basket/flywheel assembly shall be dynamically balanced and removable without tools.

3.5.1.2 Drive. The extractor shall be belt driven from the motor to a transmission. The transmission shall be a single stage fluid driven unit with a double row sealed top bearing and a single row sealed bottom bearing. An electrically operated solenoid activated braking mechanism shall be connected to the drive shaft. The brake system shall be a disk type or shoe break system which is normally on. The drive unit shall be mounted in a suspension system capable of breaking up any resonant vibration which occurs during acceleration or shock loading which occurs during road, cross country or rail travel.

3.5.1.3 Controls. The extractor shall be equipped with oil tight and water proof controls. These controls are to include but are not limited to a start button, emergency stop button and a 0 to 10 minute timer.

3.5.1.4 Signal devices. A visual indicator signal device to alert the operator that a batch is complete shall be supplied. All indicator lights, power lights, and other lighting equipment shall be covered with blue or green lenses and shall have a dimming capability. The blue or green lenses must be fabricated using an infrared light suppressant glass or plastic. Ninety-five percent of all light energy emitted between 300 and 1100 namometers shall be concentrated between 400 and 700 nanometers on any brightness setting.

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3.5.1.5 Safety devices. The extractor shall be furnished with safety devices. These devices shall include but are not limited to an automatic lid lock and a pulley and belt guard.

3.5.2 Electrical

3.5.2.1 Drive motors. The drive motor shall be constructed, classified and rated in accordance with NEMA MG 1. It shall be rated for continuous operation at 40°C or higher on 208V, 60 hertz, 3 phase power and shall be a minimum of 3 horsepower and be suitable for high speed extraction service. The motor shall have ball bearings and be totally enclosed, fan cooled, and capable of withstanding hose washdown. The motor shall be protected against overload with resettable type heaters. Reset buttons shall be accessible without opening the enclosure where the heaters are mounted. The motor shall be approved in writing for such application by the motor manufacturer.

3.5.2.2 Wiring. All wiring shall be done in accordance with NFPA No. 70 and marked in accordance with MIL-STD-195. All wiring connections shall be soldered, taped or have staked connections. The solder shall conform to type R of QQ-S-571 and the tape shall conform to HH-I-553. A permanent wiring schematic shall be permanently attached to the inside face of the control unit on a removable panel.

3.5.2.3 Enclosure. The control unit enclosure shall be equivalent to an enclosure with a type 4 rating in accordance with UL 50. All external operating mechanisms, switches, timer, and controls when mounted on or through the enclosure, shall be capable of passing the test or tests for the enclosure. All inside and outside surfaces of the enclosure shall be treated and finished as specified in 3.9. All controls and switches shall be fully accessible from the exterior of the enclosure and access to the interior of the enclosure shall not be required except to perform maintenance.

3.5.2.4 Current protection. The extractor shall be equipped with externally accessible circuit breaker protection.

3.5.2.5 Exterior lid switch. A safety interlock shall be provided to disconnect power to the drive motor when the lid is open.

3.5.2.6 Moisture and fungus-resistant treatment. Unless otherwise specified (see 6.2), electrical components shall be moisture and fungus-resistant treated with materials conforming to and applied as required in MIL-T-152.

3.6 Performance.

3.6.1 Control enclosure. The control unit enclosure shall meet the requirements of UL 50, type 4 when tested as specified in 4.6.1.

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3.6.2 Extractor lid switch (activation prevention). The extractor lid switch shall prevent operation of the drive motor when tested as specified in 4.6.2.

3.6.3 Extractor lid switch (deactivation). The extractor shall not open more than 1 inch when tested as specified in 4.6.3.

3.6.4 Emergency stop. The extractor basket shall come to a complete stop within 30 seconds when tested as specified in 4.6.4.

3.6.5 Moisture removal. The test clothes shall not retain more then 30 percent of moisture when tested as specified in 4.6.5.

3.6.6 Extractor. The extractor when operating through complete manual cycles shall not fail and show no evidence of the following conditions when tested as specified in 4.6.5 and 4.6.5.1.

a. Improper interelationship of components.
b. Improper fit, tightness or alignment of components.
c. Improper cycle times.
d. Improper operation of controls.
e. Overheated bearings or motors.
f. Leaks.
g. Structural failures.
h. Excessive vibration.
i. Excessive noise.
j. Improper motor revolutions per minute.
k. Improper basket revolutions per minute.
l. Improper braking.
m. Improper starting and running current draw.

3.6.7 Endurance test. All components shall operate properly. There shall be no damage to any of the components or any misalignment of any part. The extractor shall remove a minimum of 70 percent of the moisture from the test load and there shall be no signs of warped or distorted metal, failed or cracked welds, or any conditions outlined in 3.6.6 when tested as specified in 4.6.6 or 4.6.7 (see 6.3.1).

3.6.8 Road test. The extractor shall traverse the test course and show no damage or misalignment of any component, warped or distorted metal, failed or cracked welds, or any conditions outlined in 3.6.6. The components shall remain securely fastened to the extractor assembly with no evidence of looseness and the extractor shall remove a minimum of 70 percent of the moisture from the test load when tested as specified in 4.6.9 (see 6.3.1 and 6.3.2).

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3.6.9 Rail impact test. The extractor assembly shall be rail impact tested and show no damage to any component, warped or distorted metal, failed or cracked welds, or any conditions outlined in 3.6.6 The components shall remain securely fastened to the extractor assembly with no evidence of looseness and the extractor shall remove a minimum of 70 percent of the moisture from the test load when tested as specified in 4.6.10 (see 6.3.1 and 6.3.2).

3.7 Marking. Marking for identification shall be in accordance with MIL-STD-130.

3.8 Government furnished loan property. When the extractor is not procured as part of a contract to fabricate the Army Trailer Mounted Laundry per MIL-L-44142, an M-85 Trailer Mounted Laundry without an extractor would be government loaned property (see 6.2 and 6.3.2).

3.9 Preparation and finish of all exposed surfaces. All exterior surfaces, with the exception of glass, dataplates, hose fittings, lenses, and switches shall be prepared and finished per MIL-T-704, types F and G. The color of the finish coat shall be Green 383.

3.10 Workmanship. The units shall be free from fractures, splits, punctures, tears, dents, creases, bows, miscasts, deteriorations or malfunctions. There shall be no sharp edges, burrs or slivers. The finish shall be free of blisters, peel, chipped areas or areas of no film. The finish shall be free from smudges, poor touch up, scale, scratches, stain, overspray, sag, imprint or embedded foreign material. Fasteners shall not be broken, stripped or loose.

3.10.1 Welding. The surfaces of parts to be welded shall be free from oxide, scale, paint, grease, and other foreign matter. Welds shall be continuous, sound, smooth, and free from porosity, cracks, incomplete fusion, and deformation of material. All scale and flux deposits shall be removed from the finished welds and rough edges and projections shall be dressed smooth.

3.10.2 Electrical wiring. Wiring shall not be cut, abraded or have excessive insulation removed and shall be properly and tightly joined at terminals. Wiring shall have proper slack to provide strain relief.

3.10.3 Welders. Before assigning any welder to work, the contractor shall provide the contracting officer or the contracting officer's representative with the names of welders to be employed in the work, together with certification that each of these welders has passed a qualification test as prescribed by either of the following listed codes for the type welding operation to be performed, and that such qualification is effective as defined by that particular code as follows:

a. Standard Qualification Procedure of the American Welding Society.
b. Welding Qualification of the American Society of Mechanical Engineers.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.3 First article inspection. When a first article is required (see 3.1), it shall be examined for the defects listed in table I and paragraph 4.4.3 and, unless otherwise specified (see 6.3.1 and 6.3.2), shall be tested as specified in 4.6.1 thru 4.6.5.1, 4.6.7, 4.6.9 and 4.6.10. The presence of any defect or failure of any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be in accordance with MIL-STD-105.

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4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all of the requirements of the referenced documents unless excluded, amended, modified or qualified in this document or applicable purchase documents.

4.4.1.1 In process inspection.

4.4.1.1.1 Surface inspection. Examination of surface preparation prior to welding shall be made to determine compliance with 3.10.1. Examination of surface preparation, pretreatment and priming prior to each succeeding phase and top coating shall be made to determine compliance with MIL-T-704. Whenever nonconformance is noted, correction shall be made to the applicable process and all affected items shall be brought into compliance.

4.4.1.1.2 Wiring inspection. Examination of wiring procedures and marking shall be made to determine compliance with 3.5.2.2 and 3.10.2. Whenever nonconformance is noted, correction shall be made to the applicable process and all affected items shall be brought into compliance.

4.4.1.1.3 Moisture and fungus treatment. Examination of treatment procedures shall be made to determine compliance with 3.5.2.6. Whenever nonconformance is noted, correction shall be made to the applicable process and all affected items shall be brought into compliance.

4.4.2 End item visual examination. The end items shall be examined for the defects listed in table I, the lot size shall be expressed in units of extractors. The sample unit shall be one extractor. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Finish	Color not as specified		X
	Blistered, peeled, chipped, or area of no film.		X
	Poor touch-up, orange peel, color separation, smudge, scale, scratched, stained, overspray, sag, imprint, or foreign matter imbedded.		X

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TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Design, construction and workmanship applicable to all components and assemblies	Any component or part missing or not specified type or size	X	
	Any component that does not function, malfunctions, or is misplaced to the degree that it does not serve its intended purpose	X	
	Not fabricated as specified by the document or drawing	X	
	Not clean (evidence of oil, grease, or dirt)		X
	Sharp edge or burr		X
	Puncture, fracture, or malformed	X	
	Design detail deviating from document or drawing requirements	X	
	Evidence of scale, rust, discoloration, or deposit of any foreign matter		X
	Finish omitted or not as specified:		
	a. On brass or aluminum components		X
	b. On steel components	X	
	Area of partial or no finish		X
Riveting	Mating surfaces of riveted components not drawn up tightly	*	
	Rivet head not in contact with its mating surface	*	
	Rivet loose - not properly set	*	
	Roll of a rivet contains more than one split		X
	Portion of a rivet is set or rolled on itself in lieu of being set or rolled on the mating surface of the component	X	
Welding	Missing, incomplete, cracked, fractured, or not fused	X	
	Undercut, not smooth and uniform, or scale or flux deposits not removed		X

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Soldering (when applicable)	Missing, not adherent, or incomplete		X
	Not clean (flux residue not removed, pinholes in solder)		X
Bolts, nuts, screws, studs, and other types of threaded fasteners	Broken, stripped, fractured, or not tight		X
Electrical assembly	Wiring cut, abraded, or not properly installed or joined	X	
	Adequate slack not provided for wiring to relieve strain		X
Marking	Missing; incorrect; incomplete; illegible; of improper size, location, sequence, or method of application	X	

NOTE: Defects designated by an asterisks (*) shall be scored as "major" when seriously affecting serviceability or appearance and "minor" when affecting serviceability or appearance but not seriously.

4.4.3 End item dimensional examination. The end item shall be examined for compliance with dimensions as specified in this document and on the drawings. Any dimension that is not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of extractors. The sample unit shall be one extractor. The inspection level shall S-2 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

4.4.4 End item testing. Each production end item shall be tested for the characteristics listed in table II. Failure of any test shall be cause for rejection of the end item involved.

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TABLE II. Performance tests

Characteristic	Requirement paragraph	Test method
Extractor lid switch:		
Activation prevention	3.6.2	4.6.2
Deactivation	3.6.3	4.6.3
Emergency stop	3.6.4	4.6.4
Moisture removal	3.6.5	4.6.5
Extractor operation	3.6.6	4.6.5.1

4.4.6 Packaging examination. An examination shall be made to determine that the preservation, packaging, and marking comply with the requirements of MIL-L-3153.

4.5 Code and standard compliance. Certifications, certified test reports, or listing marks for standards submitted in accordance with 3.2 shall be examined and validated as proof of compliance. Prior to Government approval of first shipment, the contractor shall submit to the contracting officer or his authorized representative for approval, certification that the laundry extractor he proposes to furnish under this document meet the applicable requirements of NEMA MG 1, NFPA 70, and UL 50 as follows:

4.5.1 NEMA. Acceptable evidence of meeting the requirements of NEMA shall be a certificate of compliance certifying that the motors and generators conform to MG 1.

4.5.2 NFPA. Acceptable evidence of meeting the requirements of NFPA shall be a certificate of compliance certifying that the wring of the laundry extractor conforms to NFPA 70.

4.5.3 UL. Acceptable evidence of meeting the applicable requirements of the Underwriters Laboratories shall be the UL label, a UL listing mark, or a certified test report from a recognized independent testing laboratory indicating the control unit enclosure has been tested as specified and conforms to UL 50 , type 4. Such evidence must be acceptable to the contracting officer.

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4.6 Method of inspection.

4.6.1 Enclosure test. The enclosure shall be tested for the dust, atomized water, icing and hosedown performance requirements of UL 50. Any nonconformance with the requirements of 3.6.1 shall constitute failure of this test.

4.6.2 Extractor lid switch test (activation prevention). With the extractor lid open, engage the starting switch. Any nonconformance with 3.6.2 shall constitute failure of this test.

4.6.3 Extractor lid switch test (deactivation). With the extractor lid closed engage the starting switch. After 30 seconds of operation attempt to open the lid. Any nonconformance with the requirements of 3.6.3 shall constitute failure of this test.

4.6.4 Emergency stop test. Place 30 pounds (dry weight) of trousers conforming to MIL-T-44047 in water and saturate with water until they weigh 70 pounds (40 pounds of water). Place the saturated trousers into the extractor and set the timer for 4 minutes and engage the start button and extract the trousers. Operate the extractor for 2 minutes then press the emergency stop button and remove the trousers. Any nonconformance with the requirements of 3.6.4 shall constitute failure of this test.

4.6.5 Moisture removal. Place 30 pounds (dry weight) of trousers conforming to MIL-T-44047 in water and saturate with water until they weigh 70 pounds (40 pounds of water). Place the saturated trousers into the extractor, set the timer for 4 minutes and start the cycle. Remove the extracted trousers and weigh. Repeat the procedure four more times. Add and average the weights after extraction and calculate the percent moisture content. Any nonconformance with the requirements of 3.6.5 shall constitute failure of this test.

4.6.5.1 Operational test. Upon completion of the moisture removal test in 4.6.5, install appropriate test equipment and inspect the extractor for conformance to the requirements specified in 3.6.6. Any nonconformance shall constitute failure of this test.

4.6.6 Twenty hour test (see 6.3.1). Upon completion of 4.6.5.1 place, 30 pounds (dry weight) of trousers conforming to MIL-T-44047 in water and saturate with water until they weigh 70 pounds (40 pounds of water). Place saturated trousers into the extractor, set the timer for 4 minutes and start the cycle . The rate of processing shall be a minimum of 6 cycles per hour for 20 hours. Weigh the extracted trousers at the completion of each 10 extraction cycles, average the weights and calculate the percent of moisture removed. Upon completion of the test, inspect the control unit, inside of the extractor basket, the drive shaft, the drive unit, brake assembly, motor, lid and switches. Any nonconformance with the requirements of 3.6.7 shall constitute failure of this test.

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4.6.7 Endurance test. The extractor shall process trousers for 160 hours. This shall be accomplished in 8 consecutive 20 hour days. Place 30 pounds (dry weight) of trousers conforming to MIL-T-44047 in water and saturate with water until they weigh 70 pounds (40 pounds of water). Place saturated clothing into the extractor, set the timer for 4 minutes and start the cycle. The rate of processing shall be a minimum of 6 cycles per hour. Weigh the extracted trousers at the completion of the final 10 extraction cycles, average the weights and calculate the percent of moisture removed. Upon completion of the test inspect the control unit, inside of the extractor basket, the drive shaft, the drive unit, brake assembly, motor, lid and switches. Any nonconformance with the requirements of 3.6.7 shall constitute failure of this test.

4.6.8 30 mile road and terrain test (see 6.3.1). The completely assembled extractor mounted on the Government furnished Trailer Mounted Laundry shall be towed over not less then 10 miles of paved roads at speeds up to 50 mph as road and weather conditions permit, over not less then 10 miles of graveled roads at speeds averaging 35 mph (see 4.6.8.1) and over not less then 10 miles of open terrain and cross country (see 4.6.8.2). Upon completion of the road and terrain test place 30 pounds (dry weight) of trousers conforming to MIL-T-44047 in water and saturate with water until they weigh 70 pounds (40 pounds of water). Place saturated trousers into the extractor, set the timer for 4 minutes and start the cycle. Weigh the extracted trousers at the end of the cycle and calculate the percent of moisture removed. Upon completion of the test inspect the control unit, inside of the extractor basket, the drive shaft, the drive unit, brake assembly, motor, lid and switches. Any nonconformance with the requirements of 3.6.8 shall constitute failure of this test.

4.6.8.1 Paved and graveled roads. The paved and graveled road tests shall each include not less than 20 quick starts and 20 quick stops on grades of not less than 5 degrees or more than 10 degrees. Fifty percent of the starts shall be made on the ascending grades and fifty percent on the descending grades.

4.6.8.2 Open terrain and cross country. The open terrain and cross country course shall have surface irregularities consisting of mounds, rocks, brush, and irregularities normally found in cross country or open terrain. The trailer shall be pulled along the course at speeds of not less then 5 mph and not greater then 10 mph. The test drive shall include ten abrupt stops, ten abrupt starts, ten sharp left hand turns and ten sharp right hand turns.

4.6.9 Road and terrain test. The completely assembled extractor mounted on the Government furnished Trailer Mounted Laundry shall be tested by the Government at the specified location (see 6.2). The test shall be as follows:

a. 125 miles over graveled roads at speeds averaging 35 mph. The graveled road test shall include not less than 20 quick starts and stops on grades of not less than 5 percent or greater than 10 percent. Fifty percent of the starts and stops shall be made on ascending grades and 50 percent on descending grades.

b. 125 miles over Belgian block or similar course averaging 10 mph.

c. 125 miles over hilly cross-country terrain and 125 miles over level

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cross-country terrain at speeds averaging 10 mph. The open terrain and cross-country course shall have surface-irregularities consisting of mounds, rocks, brush and irregularities normally found in cross-country or open terrain. The test drive shall include 20 abrupt starts, 20 abrupt stops, 20 sharp left hand turns, and 20 sharp right hand turns.

At the completion of 500 mile road test place 30 pounds (dry weight) of trousers conforming to MIL-T-44047 in water and saturate with water until they weigh 70 pounds (40 pounds of water). Place saturated trousers into the extractor, set the timer for 4 minutes and start the cycle. Weigh the extracted trousers at the completion of the extraction cycle and calculate the percent of moisture removed. Repeat this procedure four times. Upon completion inspect the control unit, inside of the extractor basket, the drive shaft, the drive unit, brake assembly, motor, lid, and switches. Any nonconformance with the requirements of 3.6.8 shall constitute failure of this test.

4.6.10 Rail impact test. The assembled extractor shall be impact tested by the Government at a specified location (see 6.2). The test shall be in accordance with procedure IV of MIL-STD-810 and shall be as follows:

a. Forward impact at 8, 10, and 12 mph.
b. Rearward impact at 8, 10, and 12 mph.

The unit will be packaged as instructed by the manufacturer for the impact test. This packaging shall not require any special tools nor shall the time to complete the procedure, by an experienced rigger, exceed one man hour of effort. At the completion of the rail impact test, place 30 pounds (dry weight) of trousers conforming to MIL-T-44047 in water and saturate with water until they weigh 70 pounds (40 pounds of water). Place saturated trousers into the extractor, set the timer for 4 minutes and start the cycle. Upon completion inspect the control unit, inside of the extractor basket, the drive shaft, the drive unit, brake assembly, motor, lid, and switches. Any nonconformance with the requirements of 3.6.9 shall constitute failure of this test.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2) in accordance with the applicable requirements of MIL-L-3153.

5.2 Packing. Packing shall be level A, B or Commercial, as specified (see 6.2), in accordance with the applicable requirements of MIL-L-3153.

5.3 Marking. In addition to any special marking required by the contract or purchase order, unit packs and shipping containers shall be marked as specified in MIL-L-3153.

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6. NOTES

6.1 Intended use. The trailer mounted laundry extractor is to provide laundry extractor service for troops and hospitals in the field.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this document.
b. When a first article is required (see 3.1, 4.3, and 6.3).
c. When moisture and fungus-resistant treatment is not required (see 3.5.2.6).
d. Source and delivery of Government furnished loan property (see 3.8) Trailer, Mounted Laundry unit, M-85.
e. Required location of road and terrain test and rail impact test (see 4.6.9 and 4.6.10).
f. Selection for the applicable levels of preservation and packing (see 5.1 and 5.2).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.3.1 Modified first article testing for small procurements. When the procurement is for less than 25 extractors and when authorized by the contracting officer, the first article testing may be modified as follows:

a. Substitute the 20 hour extractor test specified in 4.6.6 for the 160-hour extractor test specified in 4.6.7.
b. Substitute the 30-mile road test specified in 4.6.8 for the road and terrain test specified in 4.6.9.
c. Omit the rail impact test specified in 4.6.10

The exercise of any or all of these options in no way negates the requirements in 3.6.7, 3.6.8 and 3.6.9.

6.3.2 Modified first article testing when the extractor is part of Army Laundry contract. When the extractor is procured as part of a contract for fabrication of the Army Trailer Mounted Laundry per MIL-L-44142, first article testing per 4.6.9 and 4.6.10 would be accomplished as part of the first article test for the Army Trailer Mounted Laundry.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.3).

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6.5 Subject term (key word) listing.

Laundry
Extractor

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:

Army - GL

Preparing activity:

Army - GL

Project No. 3510-A300

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-E-44166A(GL)	Extractor, Laundry: Heavy Duty Top Loading, Trailer Mounted

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP-1, Feb 86

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